INORGANIC CUME

April 15, 2011

1. (30') (a). Draw two most common types of close packing

(b). Indicate their corresponding crystal classes

(c). Point out positions and types of the interstitials (or holes)

2. (20') (a). Draw a fcc unit cell.

(b). Illustrate the lattice planes of (200), (110), and (111) in the cell.

3. (30') In CsCl crystal, Cl^- anions sit at corners and Cs^+ at the center of the cubic unit cell.

(a). What number should the ratio of r_{Cs+}/r_{Cl-} be close to?

(b). If r_{Na+} = 0.95 Å and r_{Cs+} =1.69 Å, predict the structure in NaCl crystal.

Coordination Number and Structure	Ratio of ionic radii (r+/r-)
3, Trigonal	0.155
4, Tetrahedral	0.225
6, Octahedral	0.414
8, Cubic	0.732

4. (20') Discuss and rank the solubilities of AgBr, AgCl, AgF, and AgI in water.